

COLLISION REPAIR INFORMATION BULLETIN



SUBJECT:	WELDING HIGH STRENGTH STEEL (H.S.S.)		
BULLETIN NO.:	TS–TCI–CRIB–2019	DATE ISSUED:	May, 2013
MODELS:	All Toyota & Scion Models	SECTION:	Welding

INTRODUCTION:

In today's Collision Centre environment, High Strength Steel (H.S.S.) body parts are frequently welded during collision damage repairs. Typical uses for H.S.S. are the front cross member, front body outer pillar, floor side member and door side impact protection beam. Because H.S.S. is used in these areas of potential high stress, repairs must be performed properly to maintain the structural integrity of the vehicle.

Two types of H.S.S. are used; solution hardened H.S.S. is used for door panels, hood outer panel, etc., while precipitation hardened H.S.S. is used for door impact beams and bumper reinforcements. It is not necessary to distinguish solid solution hardened steel from ordinary steels for repair purposes.

The location of H.S.S. parts can be found in the Toyota/Scion Repair Manuals for Collision Damage under the Body Panel Construction section as indicated in the sample, for the 2010-2012 Toyota Prius Collision Repair Manual on page 3 of this bulletin. For Toyota or Scion Collision Centre facilities, visit the ETAS and Toyota Technical Information System (TIS) site located in the Infostream website portal at <http://infostream.toyota.ca/> to access Repair Manuals for Collision Damage and other Collision Repair Information Bulletins. For independently operated collision centres interested in these publications, visit www.techinfo.toyota.com for a subscription and access.

REPAIR GUIDELINES:

H.S.S. and ordinary steel parts in Toyota/Scion vehicles are welded using the same methods. The following precautions apply:

- Use either spot welding or MIG/MAG (shield gas) welding. MIG/MAG (shield gas) welding should only be performed by a trained and experienced collision repair technician.
- Only braze body components previously brazed at the factory and as indicated in the Toyota/Scion Repair Manuals for Collision Damage.
- Do not use an oxy–acetylene torch for fusion welding auto bodies. The large heat affected zone may destroy galvanized coatings and cause excessive panel distortions. In addition, an oxy–acetylene torch can affect the strength and brittleness of H.S.S.
- Technicians should wear proper protective clothing or equipment and perform these operations in a safe location in order to help avoid injury.

NOTE: Issue of this Collision Repair Information Bulletin by Toyota Canada Inc. is to ensure that proper repair information is available. It does not constitute an authorization for repair. PLEASE ROUTE THIS BULLETIN TO YOUR COLLISION REPAIR CENTER MANAGER AND COLLISION & REFINISH TECHNICIANS.



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REFERENCE:

- Always follow instructions and procedures in the model specific Toyota/Scion Repair Manuals for Collision Damage when repairing collision damaged vehicles.

TOOLS:

Proper tools and equipment are mandatory for H.S.S. repair. Please see page 4 of this bulletin for a list of recommended tools and equipment that are available through the Würth Body Shop Products program to help perform the operations including:

- MIG/MAG welder
- Spot welder
- H.S.S. spot weld cutter for 8mm welds (Würth 710.0088) and 10mm welds (Würth 710.10)
- Air saw
- Plasma cutter
- Air power chisel with panel cutter
- Hole punch
- Head protector
- Dust mask (Würth 899..110,114, 100, 1)
- Face protector (Würth 984.700)
- Vise grip pliers (Würth 713.09300,09301, 09302, 09303, 09304)
- Weld through primer (see page 4 of this bulletin) (Würth 893.114113)
- Ear plugs (Würth 899.300332, 300900, 300106)
- Car cover for glass and interior (Würth 984.100)

WELDS:

Four types of welds are required in the repair of Toyota/Scion vehicles:

- Spot
- Plug
- Continuous
- Braze

Technicians performing repair work must be proficient in doing all four types of welds since all four are used in various locations throughout the vehicle. The Toyota/Scion Repair Manuals for Collision Damage show the location and type of welds that must be used. The correct number and type of welds must be used to ensure structural integrity.

PROCEDURE:

The complete in–depth procedure for welding H.S.S. and ordinary steel can be found in the Toyota/Scion Repair Manuals for Collision Damage for the applicable vehicle. The basic procedure requires:

1. Removal of auxiliary parts
2. Removal of damaged parts
3. Prepare new parts
4. Position new parts
5. Welding
6. Finishing of welded areas

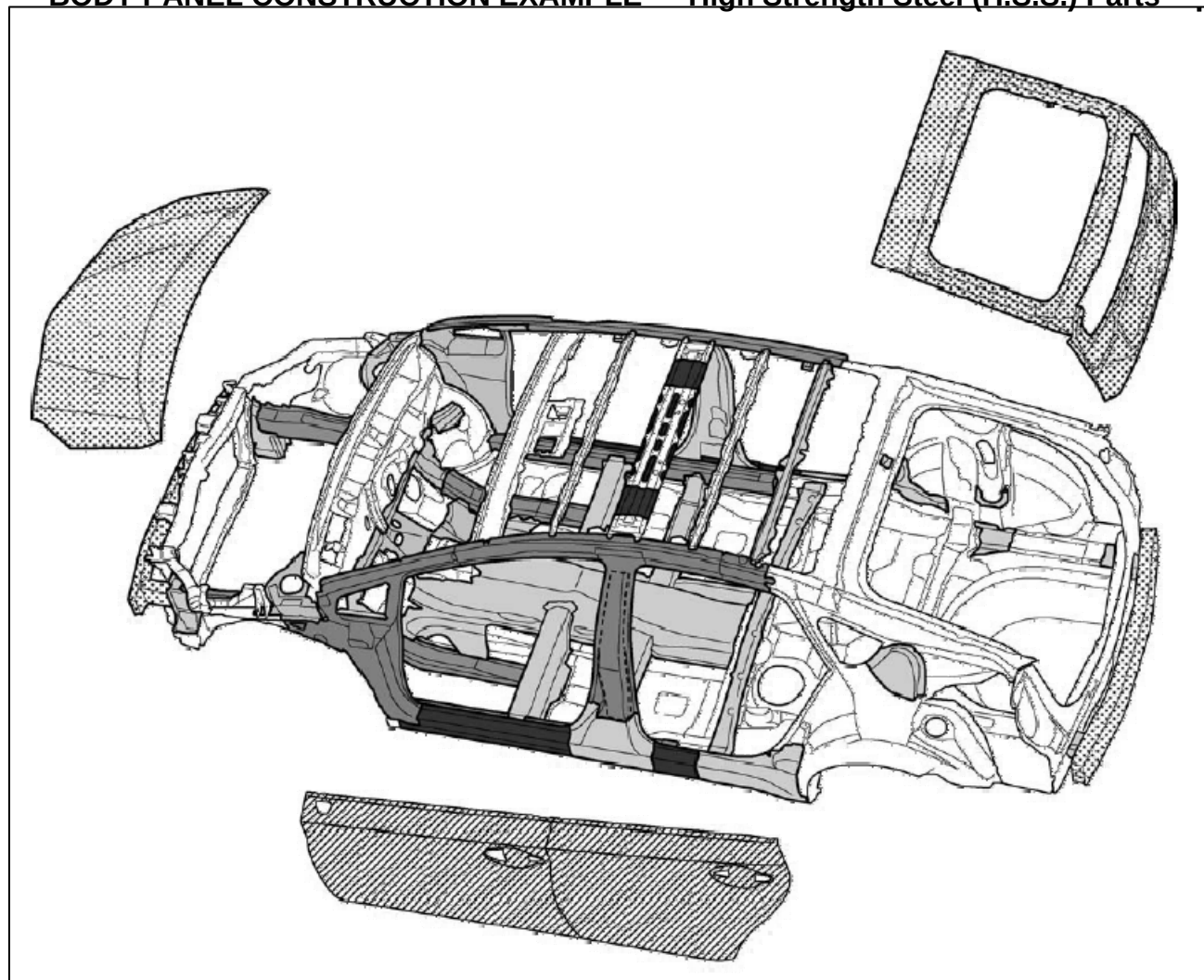


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BODY PANEL CONSTRUCTION EXAMPLE — High Strength Steel (H.S.S.) Parts



	• • • 1470 MPa Ultra high strength steel
	• • • 980 MPa Ultra high strength steel
	• • • 590 MPa High strength steel
	• • • 440 MPa High strength steel
	• • • 340 MPa High strength steel
	• • • Aluminum

2010 – 2012 Toyota Prius example shown above. Consult the appropriate Toyota/Scion manuals for actual locations for each model.



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In partnership with Würth Canada, Toyota Canada Inc. (TCI) recommends and offers a complete line-up of body shop supplies, adhesives, fasteners, abrasives, connectors, safety supplies, tools, etc., that are available through the Würth Body Shop Products Program. Toyota and Scion Dealers can order all of Würth's products through this program. Dealers can access the program details in the Infostream website portal at <http://infostream.toyota.ca/>. For independently operated collision centres interested in these products, please contact your local Toyota and Scion Dealer.

Specifically in regards to this bulletin as referenced on page 2, Toyota Canada Inc. recommends the following Würth products:

Würth Product Name	Art. No.	Illustration
Weld Through Primer	893.114113	
H.S.S. spot weld cutter for 8mm welds and 10mm welds	710.0088 & 710.10	
Dust masks	899..110,114, 100, 1	
Face protector	984.700	
Vise grip pliers	713.09300,09301, 09302, 09303, 09304	
Ear plugs	899.300332, 300900, 300106	
Car cover for glass and interior	984.100	